

Brandon Asher Woodward

Reisterstown, MD | 443-862-7644 | brandon790523@gmail.com | <https://brandonwoodward.dev/>

Software engineering student with a strong focus on real-time systems, simulation, embedded systems integration, and hardware–software co-design. Experienced in building, validating, and deploying complex systems under performance and reliability constraints. Seeking a full-time role with strong technical mentorship focused on mission-driven, real-world systems.

Education

York College of Pennsylvania, York, PA

GPA: 4.0

Bachelor of Science: Computer Science, Math Minor

Expected Graduation Date: 2026

Engineering Work Experience

Full Stack Intern | Courtside Video Entertainment, Philadelphia, PA | Summer 2025 - Present

- Designed and implemented a full-stack data-driven system using Django, React, and PostgreSQL; expanded a nontrivial relational database schema and built secure RESTful APIs.
- Built automated testing and deployment pipelines using Docker and AWS (EC2, ECR, S3, Aurora RDS), focusing on reliability and maintainability.

Engineering Intern (Part-Time) | Johnson Controls, New Freedom, PA | Spring 2025

- Automated validation of industrial chiller software, implementing 15 automated test cases and clarifying ambiguous test requirements in collaboration with documentation teams.
- Developed a LabVIEW-based hardware-in-the-loop test framework, integrating DAQs, Modbus, and RS-232, to support repeatable, one-click system testing.

Gameplay Programming Intern | Firaxis Games, Sparks, MD | Summer 2024

- Developed real-time gameplay and simulation systems in C++ using Unreal Engine 5, within a large performance-critical codebase.
- Worked on a networked multiplayer prototype, debugging synchronization, timing, and state consistency issues.
- Delivered features under strict performance, correctness, and latency constraints and presented results to senior engineers and leadership utilizing Perforce and Swarm.

Projects

Capstone: RevMetrix | York College of Pennsylvania | Spring 2025 - Fall 2026

- Developed a hardware-software-cloud system integrating Raspberry Pi, sensor modules, and motors with cloud infrastructure; coordinated system integration and cross-team execution in a 5-person team.
- Enabled arm64 support for the MetaWear library on Raspberry Pi 5 by compiling and linking multiple repositories; produced repeatable build and deployment documentation.
- Integrated BLE communication with embedded sensors; interpreted sensor datasheets to configure and collect 9-DOF IMU data, handling intermittent connectivity and ensuring system robustness.

RFID Security & Signal Analysis Project | Cybersecurity Coursework

- Used Raspberry Pi and Proxmark3 to capture and analyze RF signals emitted by iClass Legacy RFID cards.
- Developed Python and shell scripts to automate signal extraction, decoding, and replay.
- Analyzed protocol weaknesses and successfully cloned RFID credentials, demonstrating real-world security vulnerabilities in deployed access-control systems.

Leadership and Professional Development

Bronfman Fellowship: Campus Commons - Interdisciplinary Leadership & Dialogue Program

- Participated in facilitated dialogue and leadership workshops focused on navigating differences, managing tough conversations, and leading with empathy and confidence.

Gordon Leadership Scholars Program

- Selected for a competitive leadership cohort emphasizing ethical decision making, systems thinking, and collaborative problem-solving.

Additional Experience

Division III College and High School Wrestling | York College of Pennsylvania | 2018 - 2025 Spring

- Balanced multi-daily training with academics while contributing to team culture and performance.